

Calendar Component

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The calendar component displays a single-month calendar that lets the end user select dates and move to the next or previous month.

< November 2004 >						
Sun	Mon	Tue	Wed	Thu	Fri	Sat
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30				

Figure 1: Calendar Component

This document describes the calendar component and how to customize and use it in an application.

- Component objects
- Component elements
- How to customize the component
- How to attach a listener to the component
- How to use the component in an application
- How to change the look of the component

Component Objects

The calendar component holds an object of the Gregorian calendar, which is the standard calendar with 365 days and a leap year every four. The calendar object is updated each time the client selects a date. To access the Gregorian calendar object in the page bean, use the following utility method:

```
java.util.Calendar calendarObj = calendar1.getCalendar();
```

Component Elements

The calendar component contains the following individual elements:

- days of the month
- day headings for the days of the week
- title with the month name and year

- links for selecting individual days of the month
- links for moving to the next and previous month

How to Customize the Component

Customize the appearance of the calendar component by setting the style class properties (as shown in Figure 2) for any of its elements. Style and selection properties are listed and described below.

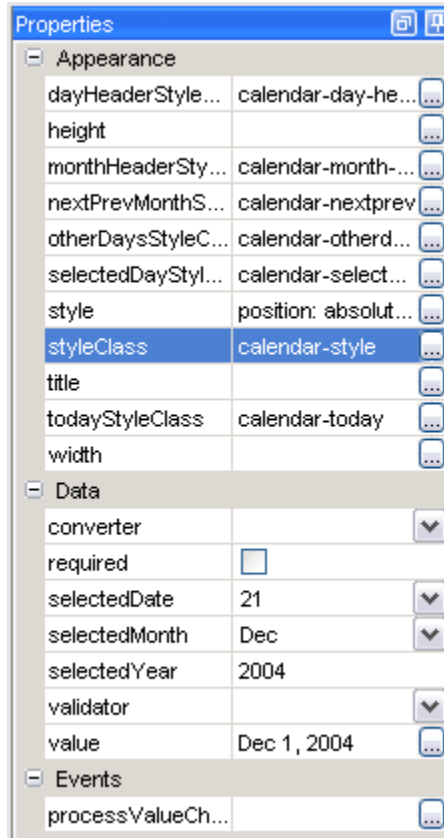


Figure 2: Calendar Properties

Style Properties

Here are the properties used to specify the style for the different parts of the component.

- MonthHeaderStyleClass** Specifies the style for the section that displays the name of the month.
- DayHeaderStyleClass** Specifies the style for the section that displays the days of the week.
- NextPrevMonthStyleClass** Specifies the style for the month navigation controls in the title section.
- OtherDaysStyleClass** Specifies the style for the dates that are not today's date in the currently displayed month.
- SelectedDayStyleClass** Specifies the style for the selected dates on the calendar.
- TodayDayStyleClass** Specifies the style for today's date.

Selection Data Properties

Here are the properties used to set or get the selection data for the component.

selectedDate	Sets or gets the day of the month of the component.
selectedMonth	Sets or gets the month the component.
selectedYear	Sets or gets the Year of the component.
value	Returns a String representation of the selected date.

How to Attach a Listener to the Component

Double click on the component to automatically attach an event listener to it. The following code will appear in the JSP and page bean:

In the JSP:

```
<a:calendar binding="#{Page1.calendar1}"  
valueChangeListener="#{Page1.calendar1_processValueChange}"
```

In the page bean:

```
public void calendar1_processValueChange(ValueChangeEvent vce) {  
    // User code goes here  
}
```

When the user clicks the link to any of day of the month in the calendar component, the page is submitted and the value change listener is invoked.

How to Use the Calendar Component in an Application

To use the calendar component in an application first open Java Studio Creator to view the list of available components in the JSF standard components palette.

1. Drag and drop the calendar component onto the Java Studio Creator design canvas.
2. Drag and drop four text components on to the design canvas and set their values as follows (shown in Figure 3).
 - outputText1 value as "Selected Date: (default)"
 - outputText3 value as "Selected Date: (formatted)"
 - Do not change the values of outputText2 and outputText4 (populated at run time)

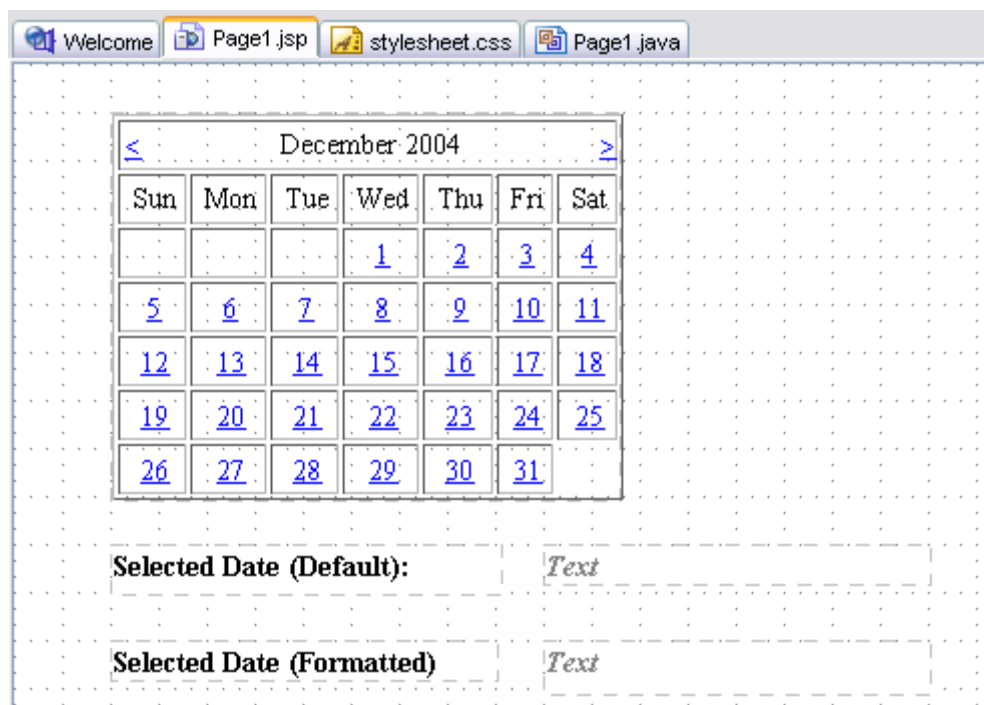


Figure 3: Calendar component on the visual design canvas

3. Double click on the calendar component to access the Java source. Notice that the event listener has been automatically created. Enter the following code:

```
public void calendar1_processValueChange(ValueChangeEvent vce) {
    outputText2.setValue(this.calendar1.getValue());
    Calendar calen = calendar1.getCalendar();
    DateFormat dateFormat = new SimpleDateFormat("EEE, d MMM yyyy HH:mm:ss Z");
    outputText4.setValue(dateFormat.format(calen.getTime()));
}
```

4. Build and run the application that contains the calendar component.
5. In the browser, click to select any date. The result should be as shown in Figure 4, below:

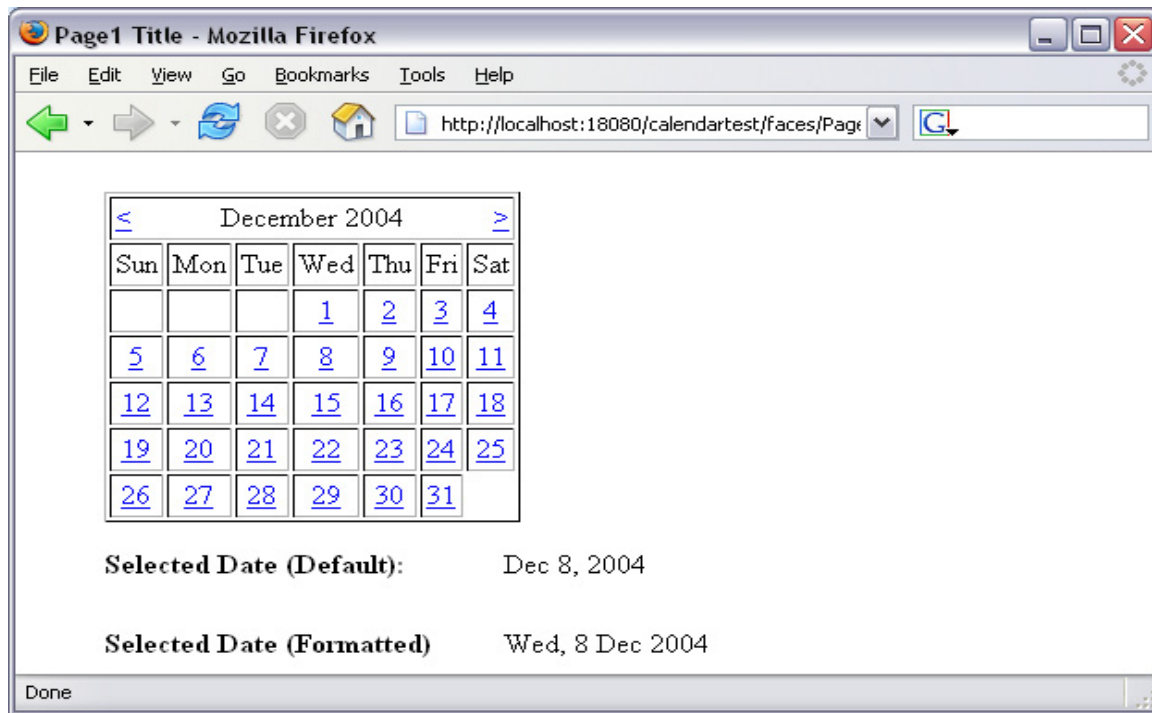


Figure 4: Calendar component in the Browser

How to Change the Look of the Calendar

The various style class attributes of the calendar component can be modified to vary background colors and fonts from the default white background and font shown in Figure 1. Here's how to make these changes.

1. Open the CSS file `resources/stylesheet.css` from the project node. Add the following CSS style classes to the stylesheet to create the look shown in Figure 5.

```
.calendar-header {
    background-color: #DDDDFF;
}
.calendar-month-header {
    font-size: 12pt;
    font-weight: bold;
    background-color: rgb(183,183,255);
}
.calendar-day-header{
    color: blue;
    background-color: #CCCCCC;
}
.calendar-selected-day{
    background-color:#AAAAFF;
}
.calendar-today {
    font-weight: bold;
    background-color: #FFAAAA;
}
.calendar-otherdays {
    background-color: #FFFFCC;
}

.calendar-nextprev {
    background-color: rgb(183,183,255);
}
```

<	December 2004						>
<i>Sun</i>	<i>Mon</i>	<i>Tue</i>	<i>Wed</i>	<i>Thu</i>	<i>Fri</i>	<i>Sat</i>	
			1	2	3	4	
5	6	7	8	9	10	11	
12	13	14	15	16	17	18	
19	20	21	22	23	24	25	
26	27	28	29	30	31		

Figure 5: Color and font changes

2. In the design palette, select the calendar component to display its property sheet. Invoke the custom editor of various style classes from the property sheet shown in Figure 6.

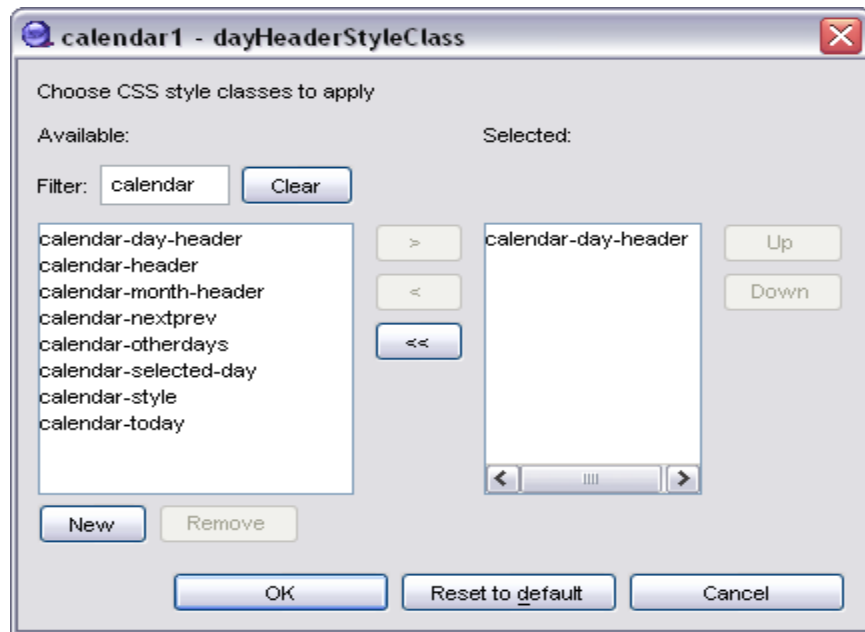


Figure 6: Style class selection dialog

3. Choose the style classes to apply to the calendar component from the `Style Class selector` dialog.

Calendar Attributes

MonthHeaderStyleClass

DayHeaderStyleClass

NextPrevMonthStyleClass

OtherDaysStyleClass

SelectedDayStyleClass

TodayDayStyleClass

Style classes

Calendar-month-header

calendar-day-header

calendar-nextprev

calendar-otherdays

calendar-selected-day

calendar-today

For more information and documentation on components and their use please visit the Java Studio Creator [website](#).

About the Author

Dr. Winston Prakash is a member of the Sun Java Studio Creator development team. He has been developing enterprise software for the past 15 years with a focus on programming tools for J2EE™ developers since 2000.